

SEABAD: A Tropical Bird Activity Detection Dataset for Passive Acoustic Monitoring

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Dataset: <https://zenodo.org/records/18290494> **Code:** <https://github.com/mun3im/seabad>

Abstract

Passive acoustic monitoring (PAM) enables large-scale biodiversity assessment, but continuous recording generates large amounts of non-informative audio, creating challenges for storage, power consumption, and long-term edge deployment. Bird audio detection (BAD), which identifies bird vocalizations, can reduce this burden by filtering irrelevant recordings before downstream analysis. However, most BAD systems are trained on temperate datasets despite tropical soundscapes being denser, more species-rich, and acoustically unpredictable.

To address this gap, we introduce SEABAD (Southeast Asian Bird Activity Detection), a dataset of 50,000 curated three-second clips from Southeast Asian soundscapes, evenly balanced between bird-present and bird-absent samples. The dataset spans 1,677 bird species and is standardized to 16 kHz mono audio for embedded and low-power inference. We developed a dual-branch curation pipeline: a six-stage positive-label workflow applied to Xeno-Canto recordings, alongside six source-specific negative-label extractions from environmental datasets. These procedures reduced class imbalance by 13.7% (Gini coefficient: 0.601 $\rightarrow$ 0.519). A manual audit of 1,000 positive clips confirmed $97.8\% \pm 0.9\%$ labeling accuracy. Baseline experiments using MobileNetV3-Small achieved $99.57\% \pm 0.25\%$ accuracy and 0.9985 ± 0.0002 AUC across three random seeds. SEABAD and the full curation pipeline are publicly released to support tropical BAD research and energy-efficient acoustic monitoring.

Keywords: passive acoustic monitoring; bird activity detection; edge AI; tropical soundscapes; bioacoustics; dataset curation; biodiversity informatics

1 Introduction

Tropical ecosystems support more than half of the world's terrestrial biodiversity, yet they are also among the regions undergoing the fastest ecological change and habitat degradation. Because birds respond sensitively to shifts in vegetation structure, habitat quality, and ecosystem stability, their vocal activity is widely used as an indicator of broader environmental conditions (Fraixedas et al. 2020; Sethi et al. 2022). This has made long-term avian monitoring increasingly important for conservation planning, ecological assessment, and biodiversity research.

Traditional survey methods such as point counts and transect observations remain valuable, but they are difficult to scale

and depend heavily on observer availability and expertise (Chauvier et al. 2021; Pérez-Granados and Traba 2021). These challenges become even more pronounced in tropical forests, where dense canopy cover, uneven terrain, and exceptionally high species diversity complicate direct observation (Mendoza-Henao et al. 2023; Crunchant et al. 2021; Gibbons et al. 2026). In response, passive acoustic monitoring (PAM) has emerged as a widely adopted alternative. By continuously recording environmental soundscapes over long periods, PAM enables non-invasive monitoring across locations and timescales that would be difficult to survey manually (Priyadarshani et al. 2018; Melo et al. 2021; Ross et al. 2023).

The practical problem is that most of these recordings contain very little useful information. Autonomous recording units

(ARUs) are commonly configured to record at fixed intervals or respond to simple amplitude thresholds, which means large portions of the collected audio consist of wind, rainfall, insects, or other background noise rather than bird vocalizations. In some deployments, bird activity accounts for less than 10% of the recorded material (Huang et al. 2024). For field systems operating under tight storage and energy budgets, this inefficiency reduces deployment duration and substantially increases the effort required for downstream processing and annotation.

Recent progress in TinyML has made it possible to run neural inference directly on low-power microcontroller platforms (Banbury et al. 2021; David et al. 2021). This creates an opportunity to move beyond continuous indiscriminate recording toward more selective acoustic sensing. Within this setting, Bird Activity Detection (BAD) serves as a lightweight front-end task that determines whether bird vocalizations are present in an audio segment. Non-informative recordings can then be discarded before storage or transmission. Compared with full species classification, binary presence-absence detection can be implemented using relatively compact models that fit within the memory and energy constraints of devices such as AudioMoth recorders, making BAD a practical strategy for improving PAM efficiency (Stowell and Plumbley 2014; Stowell et al. 2018; da Silva et al. 2019; Lostanlen et al. 2019).

However, progress in low-power BAD is constrained less by model architecture than by data availability and dataset construction practices. Most publicly available bird audio datasets originate from temperate regions, and models trained on these corpora often generalize poorly to tropical soundscapes (Burivalova et al. 2022). Tropical environments present distinct acoustic conditions, including high species richness, frequent multi-species choruses, and dense non-avian biophony from sources such as cicadas and primates. Together, these factors alter soundscape statistics and reduce detection performance when models are transferred from temperate to tropical domains. Existing datasets are also rarely designed for embedded presence-absence detection, and few studies provide reproducible pipelines for building region-specific BAD datasets.

Current datasets therefore do not fully satisfy the requirements for edge-deployable BAD in Southeast Asia. In particular, few offer short fixed-duration clips compatible with microcontroller buffer constraints (3 s at 16 kHz), binary presence-absence labels rather than species-level annotations, and broad representation of Southeast Asian tropical soundscapes. These requirements are seldom addressed together within a single dataset, creating a mismatch between embedded inference constraints and the acoustic complexity of tropical ecosystems.

To address this gap, we introduce **SEABAD (Southeast Asian Bird Activity Detection)**, a large-scale tropical dataset and a reproducible framework for constructing bird presence-absence corpora. SEABAD consists of 50,000 automatically curated 3-second clips spanning more than 1,600 species across Southeast Asian ecosystems, standardized

to 16 kHz mono for embedded deployment. Alongside the dataset, we present an end-to-end curation pipeline covering data acquisition, automated segment extraction, acoustic deduplication, and diversity-aware species balancing.

Contributions.

1. **A tropical BAD dataset:** SEABAD, a 50,000-clip dataset spanning 1,677 Southeast Asian species, explicitly designed for edge-deployable bird presence detection.
2. **A reproducible dual-branch pipeline:** A six-stage positive-label workflow (metadata acquisition through quality assurance) running in parallel with six source-specific negative-label extractions, jointly producing 50,000 clips.
3. **A diversity-aware balancing method:** A strategy that reduces long-tailed species imbalance by 13.7% (Gini coefficient) while preserving intra-species acoustic variability.
4. **Strong baseline performance:** Lightweight CNNs achieve >99% AUC, indicating high label quality and task separability.

2 Related work

2.1 Hardware and sensing infrastructure

While proprietary systems like Wildlife Acoustics Song Meter dominate conservation deployments (Pérez-Granados et al. 2019; Mennill 2024), their cost limits accessibility. Open-source alternatives like AudioMoth (Hill et al. 2019; Bota et al. 2023) combine low-cost design with energy-efficient recording using Goertzel filters for frequency-band monitoring. Recent advances enable on-board neural inference on AudioMoth’s Cortex-M4 processor (Ciapponi et al. 2025), marking a shift toward edge-based acoustic intelligence.

2.2 Bird activity detection: algorithms and datasets

Bioacoustic analysis has shifted from feature-based methods (MFCCs with SVMs; Mutanu et al. 2022) to deep CNNs that learn spectro-temporal representations (Stowell 2022). Most research focuses on species classification (e.g., BirdNET; Kahl et al. 2021b), while BAD or binary presence detection remains underexplored despite its importance for edge preprocessing.

Early BAD methods used spectrogram segmentation (Potamitis et al. 2014) and handcrafted acoustic features (Zhao et al. 2017), requiring manual feature engineering. The DCASE 2018 challenge (Stowell et al. 2018) standardized BAD benchmarking with Freefield1010, Warblr, and BirdVox-DCASE datasets (10-second clips), but introduced weak labeling (bird presence <5% of clip) and temporal mismatch with modern 3-second architectures, complicating edge deployment (Solomes and Stowell 2020). Follow-on work demonstrated that training across mixed-source datasets improves generalization to unseen recording conditions (Kim

et al. 2020), motivating the heterogeneous negative sample pool used in SEABAD.

Recent studies have explored finer-grained localization using temporal segmentation (Cohen et al. 2022) and patch-based CNN architectures (Lostanlen et al. 2024). Although effective, these methods depend on dense convolutional processing, making them difficult to run on highly constrained hardware such as the AudioMoth platform, which operates within 256 kB of RAM and strict real-time latency limits of roughly 100 ms. Frameworks designed specifically for embedded BAD, such as TinyChirp (Huang et al. 2024), show that careful optimization can make edge deployment feasible, but the trade-off between computational cost and detection reliability remains an open challenge.

2.3 Geographic bias in bioacoustic datasets

Global biodiversity data remain heavily imbalanced. More than 70% of occurrence records come from Europe and North America, even though these regions account for less than one-third of the Earth’s land surface (Hughes et al. 2021). The same pattern appears in bioacoustic datasets, where recording efforts and sensor deployments are concentrated in temperate regions. As a result, models trained on these datasets often perform poorly in tropical soundscapes (Stowell et al. 2019; Winiarska et al. 2024).

Tropical environments differ substantially in their acoustic structure, with higher species richness, overlapping multi-species choruses, and dense non-avian biophony from insects and primates. These differences make it difficult for models trained in temperate regions to generalize effectively, highlighting the need for region-specific datasets (Jeliazkov et al. 2022; Meyer et al. 2016).

2.4 Dataset annotation strategies

One of the main differences between bird audio datasets is how the recordings are annotated. Broadly, existing datasets tend to follow three approaches. Some rely on **crowdsourced soft labels**, where contributors indicate only whether a species is present or absent, as seen in Freefield1010 and Warblr. Others use **expert hard labels**, where trained annotators mark precise temporal boundaries for vocalizations, as in BirdVox-DCASE. A third category combines automatic labeling with partial manual verification, as in BirdSet. Table 1 compares SEABAD with representative datasets.

SEABAD follows a hybrid strategy. Positive samples inherit species labels from crowd-sourced metadata available in Xeno-Canto, after which recordings are segmented automatically using RMS-based energy detection to extract fixed 3-second regions centered on periods of strong vocal activity. This design retains the scalability of community-generated archives while introducing temporal consistency suitable for detection tasks.

Recordings with Xeno-Canto quality ratings of A or B are strongly prioritized during selection — 92.1% of the final dataset carries these ratings — and the segmentation process

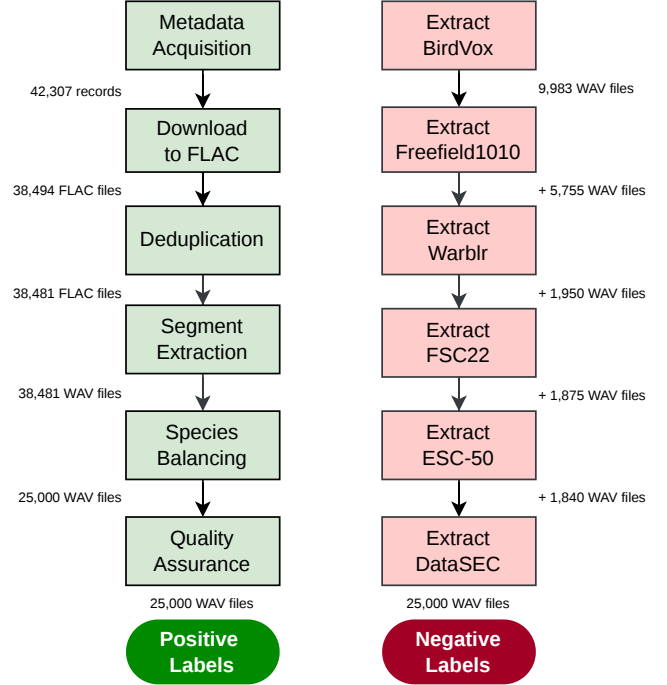


Figure 1: Dual-branch curation pipeline used to construct SEABAD. The positive-label branch (left) processes Xeno-Canto recordings from five Southeast Asian countries through six sequential stages: metadata acquisition, download to FLAC, acoustic deduplication, segment extraction, diversity-aware species balancing, and quality assurance, yielding 25,000 bird-present clips. The negative-label branch (right) runs six parallel source-specific extractions from BirdVox-DCASE-20k, Freefield1010, Warblr10k, FSC-22, ESC-50, and DataSEC, yielding 25,000 bird-absent clips. Both branches produce 3-second, 16 kHz mono WAV files; the combined 50,000-clip dataset is used for downstream validation (not shown).

preferentially selects regions with the highest acoustic energy, which are more likely to contain the intended vocalization.

3 Materials and methods

The dataset curation procedure (Figure 1) consists of preparing positive (bird present) and negative (bird absent) labels. For positive labels, sounds are sourced from the Xeno-Canto open-access repository and processed using the procedure described below. For negative labels, non-bird sounds are sourced from the BirdCLEF dataset (Freefield1010, Warblr, and BirdVox), ESC-50 (Environmental Sound Classification), and FSC-22 (Forest Sound Classification).

3.1 Positive label curation

Curating high-quality, ecologically diverse training data from community repositories such as Xeno-Canto poses several challenges. Raw downloads exhibit (1) acoustic duplicates due to re-uploads and cross-border species, (2) severe class imbalance with a long-tailed species distribution, and (3)

Table 1: Comparison of bird audio datasets for detection and classification tasks. SEABAD uses crowd-sourced soft-labeling and automatic hard-labeling/segmentation.

Dataset	Region	Task	Clips	Species	Clip length	Annotation
Freefield1010 (Stowell and Plumbley 2014)	Global	Detection	7,690	unlabeled	10 s	Crowd
Warblr (Stowell et al. 2018)	UK	Detection	10,000	20+	10 s	Crowd
BirdVox-DCASE (Lostanlen et al. 2018)	N. America	Detection	20,000	<10	10 s	Expert
BirdSet (Rauch et al. 2025)	Global	Classification	520,000+	10,000+	variable	Mixed
LifeCLEF (Joly et al. 2018)	Global	Classification	large	1,000+	variable	Mixed
SEABAD	SE Asia	Detection	50,000	1,600+	3 s	Mixed

limited intra-species diversity caused by repeated use of the same recordings.

To address these issues, we developed a six-stage positive-label curation pipeline featuring novel algorithms for acoustic deduplication and diversity-aware species balancing. The positive-label branch processes Xeno-Canto recordings through the following stages:

1. metadata acquisition,
2. download and conversion to FLAC,
3. acoustic deduplication,
4. segment extraction,
5. species balancing, and
6. quality assurance.

3.1.1 Metadata acquisition

Species metadata and recording information were retrieved using the Xeno-Canto API (v3), which provides a unique identifier, species taxonomy, recording quality rating, geographic coordinates, and download links for each recording. All taxa cataloged as birds were included, with coverage determined entirely by recording availability on Xeno-Canto.

SEABAD focuses on the Sundaland biogeographic region and southern Indochina, comprising Malaysia, Indonesia, Singapore, Brunei, and Thailand. These countries share substantial avifaunal overlap resulting from historical Pleistocene land-bridge connectivity and collectively represent approximately 1,677 bird species. This regional focus captures a large portion of Southeast Asia’s tropical lowland avifauna while maintaining ecological and biogeographic coherence. Recordings from additional Southeast Asian countries (e.g., the Philippines, Vietnam, Laos, Cambodia, and Myanmar) were available but excluded from this release to maintain geographic consistency. The curation pipeline is designed to support future expansion to a broader pan-Southeast Asian dataset.

The API query returned 43,108 records labelled as `grp = "birds"`. After removing 801 entries with unresolved species identities (e.g., “unknown” or “identity unknown”), 42,307 records with valid species labels were retained.

3.1.2 Sound acquisition and resampling

Only entries with valid and accessible audio URLs were downloaded. Recordings that were blocked, unavailable,

or shorter than 3 seconds were excluded. All audio files (downloaded from Xeno-Canto in MP3 format) were converted to mono 16 kHz FLAC using `ffmpeg` with polyphase anti-aliasing resampling for lossless intermediate processing. Preprocessing was deliberately minimal — limited to resampling and format conversion — to emulate real edge recording conditions. No amplitude normalization or signal conditioning was applied.

This procedure yielded 38,494 usable recordings. A total of 3,799 entries were excluded due to missing URLs or insufficient duration, and 14 downloads failed to convert after download.

3.1.3 Acoustic similarity-based deduplication

Community-contributed repositories can contain duplicate recordings due to re-uploads, metadata revisions, or format conversions. Such duplicates pose a risk of train-test contamination when they span data splits. Traditional hash-based methods fail to detect acoustically identical files stored with different codecs or bitrates. We therefore implemented acoustic similarity-based deduplication using mel-spectrogram embeddings and FAISS (Johnson et al. 2019) approximate nearest-neighbor search.

Each 3-second recording was converted to a 128-bin mel spectrogram (16 kHz, 512-point FFT, 128-sample hop). A compact 256-dimensional embedding $\mathbf{z} = [\mu(\mathbf{E}), \sigma(\mathbf{E})]$ was computed by concatenating mean and standard deviation across time frames, then L2-normalized for cosine similarity comparison. FAISS indexed all candidate embeddings, retrieving the top-6 nearest neighbors per clip, avoiding $O(N^2)$ exhaustive comparison.

Duplicate pairs were identified when embeddings were identical within machine precision ($\|\mathbf{z}_i - \mathbf{z}_j\|_2 < 10^{-7}$), which occurs when two clips produce numerically indistinguishable mel-spectrogram statistics. This conservative criterion avoids false duplicate removal while still detecting exact acoustic duplicates. For each duplicate pair, the file with the higher Xeno-Canto catalog number was removed, preserving earlier uploads.

3.1.4 Segment extraction

Recordings were segmented into fixed-length 3 s clips to capture complete vocalizations while maintaining computational efficiency. Each recording was analyzed using

a sliding window (3 s window, 100 ms step) with RMS amplitude computed for each position:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} y[n]^2},$$

where $y[n]$ denotes waveform samples and N the window length. Windows with $\text{RMS} \geq 0.001$ were ranked by energy, and clips selected while enforcing 1.5 s minimum temporal separation to prevent redundancy. For recordings > 12 s, the first 3 s were skipped to avoid voice annotations or handling noise. Clips with amplitude clipping (peak ≥ 0.9999) were corrected using peak scaling and soft limiting. Full-retention mode preserved all extracted clips for subsequent species-level balancing.

3.1.5 Species balancing

Problem formulation Community-contributed archives exhibit long-tail species distributions where common species dominate (Stowell et al. 2018; Kahl et al. 2021a). This biases models toward overrepresented species, reducing performance on rare taxa critical for biodiversity monitoring (He and Garcia 2009; Priyadarshani et al. 2018; Benkendorf et al. 2023).

After deduplication, 38,481 clips spanning 1,677 species showed high inequality (mean 22.9 clips/species, Gini 0.601). Balancing required reducing this inequality while preserving acoustic diversity within species and recording-level diversity across sources.

Related balancing strategies Several strategies have been proposed to address imbalance in bioacoustic datasets. In general machine learning, random undersampling and synthetic oversampling techniques such as SMOTE are commonly used (He and Garcia 2009; Chawla et al. 2002). However, SMOTE-style synthesis is unsuitable for ecological datasets because artificial vocalizations would not correspond to real biological recordings.

Within bioacoustics, balancing is typically achieved through simple per-species caps, as widely practiced in BirdCLEF challenge pipelines (Joly et al. 2018; Kahl et al. 2021a). BirdVox-70k (Lostanlen et al. 2018) limits clip density per recording to prevent near-duplicate segments. BirdSet (Rauch et al. 2025) applies quality filtering and multi-stage curation analogous to our approach. NIPS4Bplus (Morfi and Stowell 2018) accepts class imbalance to reflect natural soundscapes. Our approach extends these practices by combining recording-level diversity, large-scale Xeno-Canto curation, and ecological long-tail awareness with explicit acoustic clustering to ensure selected samples represent diverse call types within each species.

Acoustic diversity-aware undersampling We implemented an acoustic diversity-aware stratified undersampling procedure that combines species-level balancing with within-species acoustic clustering and quality-aware selection. The algorithm operates in three stages.

First, an *acoustic salience score* is computed for each clip:

$$\text{salience} = 0.7 \cdot \frac{\text{mean_contrast}}{40.0} + 0.3 \cdot \frac{\text{mean_centroid}}{f_s} \quad (1)$$

This score prioritizes clips with clear foreground vocalizations and penalizes low-energy or noisy segments.

Second, clips for each species are grouped using MiniBatch K-Means clustering applied to mel-spectrogram embeddings into five acoustic clusters representing distinct vocal behaviors (e.g., songs, calls, or different call types). Given a target dataset size N_{target} and S species, the base per-species allocation is:

$$n_{\text{base}} = \left\lfloor \frac{N_{\text{target}}}{S} \right\rfloor \quad (2)$$

For species with fewer than n_{base} clips, all samples are retained. For species exceeding this threshold, representative clips are selected across acoustic clusters, prioritizing those with higher salience.

Third, a priority-queue backfilling stage expands the dataset to the target size by selecting additional clips ranked by:

$$\text{score} = \text{salience} + \text{quality_bonus} + \text{diversity_bonus} \quad (3)$$

where quality bonuses favor A/B-rated recordings and diversity bonuses prioritize clips from previously unused acoustic clusters. If the resulting set exceeds N_{target} , a final global trimming step retains the highest-salience clips.

Gini coefficient To quantify dataset inequality, we compute the Gini coefficient (Gini 1909; Giorgi and Gigliarano 2016):

$$G = \frac{2 \sum_{i=1}^S i n_i}{S \sum_{i=1}^S n_i} - \frac{S+1}{S} \quad (4)$$

where n_i denotes the sample count of the i -th species sorted in ascending order. Values range from 0 (perfect equality) to 1 (maximum inequality). This metric is widely used to measure ecological evenness and class imbalance in biodiversity datasets (Wittebolle et al. 2009; Daly et al. 2018).

3.2 Negative label curation

Robust BAD requires carefully curated negative examples that capture the acoustic complexity of natural environments. False positives from weather events, insect choruses, machinery, and human activity can severely degrade field performance, particularly in tropical deployments where soundscapes exhibit high temporal and spectral variability. We assembled 25,000 bird-absent clips from multiple open-access datasets (Table 2), ensuring both acoustic diversity and ecological realism.

Table 2: Public sound datasets used to extract non-bird samples.

Dataset	Duration	Bird classes	Non-bird classes	Bird samples	Non-bird samples	Usable samples	Geography
BirdVox-DCASE-20k	10 s	1	1	10,017	9,983	9,983	NE USA
Freefield1010	10 s	1	1	1,935	5,755	5,755	Global
Warblr	10 s	1	1	6,045	1,955	1,950	UK
FSC-22	5 s	3	27	150	1,875	1,875	European forests
ESC-50	5 s	4	46	160	1,840	1,840	Global
DataSEC	varies	3	19	233	4,059	3,597	Southern Europe
Total negative samples						25,000	

3.2.1 Curation strategy and preprocessing

DCASE bird detection datasets The foundation of our negative samples derives from DCASE 2018 bird detection benchmarks: BirdVox-DCASE-20k (Lostanlen et al. 2018), Freefield1010 (Stowell and Plumbley 2014), and Warblr10k (Stowell et al. 2018). These datasets provide expert-verified *hasbird*=0 annotations from realistic outdoor conditions — wind, rain, insects, distant anthropogenic sounds, and ambient noise — precisely the acoustic events that commonly trigger false alarms in automated detection systems (Stowell et al. 2018; DCASE Community 2018).

The DCASE 2018 task comprises three development sets (Freefield1010 with 7,690 clips, Warblr10k with 8,000 clips, and BirdVox-DCASE-20k with 20,000 clips) plus three evaluation sets. From these, we extracted 17,685 verified negative examples. DCASE’s 10-second clip format introduces a temporal mismatch with modern 3-second architectures; we reprocessed all recordings into centered 3-second segments to harmonize temporal resolution and align with edge-AI deployment requirements.

FSC-22 forest sound classification The FSC-22 dataset (Bandara et al. 2023) was used to add ecologically realistic non-avian biological sounds from both tropical and temperate forest environments. After removing avian categories (BirdChirping, WingFlapping), 1,875 samples containing mammals, amphibians, insects, rainfall, and wind were retained. These sounds are commonly present in forest monitoring recordings but are often underrepresented in bird detection benchmarks.

ESC-50 environmental sounds ESC-50 (Piczak 2015; Fonseca et al. 2017) provides 2,000 environmental recordings across 50 semantic classes, contributing urban, mechanical, and human-made distractors such as footsteps, engines, doors, and machinery. After excluding avian classes (chirping_birds, crow, rooster, hen), 1,840 samples remained.

DataSEC Mediterranean soundscapes The DataSEC dataset (Bandara et al. 2023) provides recordings from Mediterranean agricultural and rural environments in southern Europe. After excluding avian-related categories (Birds, Chicken coop, Crows seagulls and magpies), the remaining recordings include vehicles, aircraft, machinery,

sirens, thunder, bells, insects, cats, dogs, and human speech. Recordings shorter than 3 s were zero-padded to the target duration rather than discarded; retaining these short clips provided sufficient samples to meet the per-source quota without drawing from the music category, which was excluded from the corpus. Longer recordings were segmented by selecting the 3-second window with highest RMS energy. This produced 3,597 Mediterranean soundscape clips.

3.2.2 Preprocessing pipeline

All negative samples underwent standardized preprocessing:

1. **Resampling:** All recordings resampled to 16 kHz mono.
2. **Segmentation:** Clips trimmed or centered to exactly 3 seconds.
3. **Dynamic range:** No compression applied; clipping artifacts were minimal across datasets.
4. **Metadata logging:** Each clip cataloged with source file, dataset origin, start timestamp, and RMS amplitude in a unified CSV.

3.2.3 Quality filtering

Phase 1: Quality filtering. We applied strict quality criteria to 3-second center clips:

$$\begin{aligned} \text{RMS} &\geq 0.0001 \\ \text{Peak amplitude} &\leq 0.98 \\ \text{Dynamic range} &\geq 0.1 \end{aligned}$$

These thresholds remove silent clips, clipped or distorted audio, and signals with insufficient variation.

Phase 2: Deployment-aware allocation. For multi-category datasets (ESC-50, FSC-22, DataSEC), avian classes were first removed to preserve negative-sample purity. Remaining categories were prioritised by ecological relevance: environmental and outdoor recordings were selected first, followed by voice and human-activity sounds. Recordings shorter than 3 s were zero-padded rather than discarded, as this increased the yield of ecologically realistic samples and avoided the need to draw from the music category, which was excluded from the final corpus as its structured harmonic content is rarely encountered in field PAM deployments.

Phase 3: Diversity maximization. For retained categories with large clip counts (e.g., voices in DataSEC), a random subset was drawn without replacement to cap per-category contribution and prevent any single sound type from biasing the trained model.

3.3 Quality assurance

3.3.1 Audit procedure

To assess the labeling accuracy of the automated curation pipeline, two independent random samples totaling 1,000 positive clips were inspected by the first author using mel-spectrogram visualizations and audio playback (Figure 2). Each audit round presented 500 clips as 20 pages of 5×5 spectrogram grids (25 clips per page) rendered at 4K resolution, enabling efficient batch inspection while preserving visual detail. Each clip was assigned one of four outcomes: *correct* (salient bird vocalization visible and audible), *wrong onset* (vocalization present in the source recording but outside the extracted 3 s window), *noise dominated* (broadband environmental noise masks the vocalization), or *no bird* (no vocalization detectable).

Clips affected by onset errors were remediated using an interactive correction tool that displays the source FLAC spectrogram and waveform, allowing manual adjustment of the 3 s extraction window. Corrected onset times were recorded in a structured log, then re-extracted programmatically to ensure reproducibility. Clips containing no detectable vocalization ($n = 1$) were removed and replaced by re-running segment extraction on unused regions of the same source recordings.

Negative clips inherit their labels from the original source datasets (DCASE, ESC-50, FSC-22, DataSEC), which were independently verified by their respective authors. No additional manual audit of negative clips was performed beyond the RMS and peak-amplitude quality filters.

3.3.2 Audit sample size

The required audit sample size was determined using Cochran’s formula for estimating proportions in finite populations (Cochran 1977). Using the observed error rate from the first audit round ($\hat{p} = 0.04$), a target margin of error $e^* = 0.015$ (± 1.5 percentage points), and 95% confidence ($z = 1.96$), the required sample size is:

$$n_0 = \frac{z^2 \hat{p}(1 - \hat{p})}{e^{*2}} = 656, \quad n^* = \frac{n_0}{1 + n_0/N} = 639, \quad (5)$$

where $N = 25,000$ is the population size. A second audit round of 500 clips (total $n = 1,000 > 639$) was therefore conducted. This sampling strategy follows established practice in machine learning dataset quality auditing (Northcutt et al. 2021).

3.4 Baseline validation experimental setup

To validate dataset quality, four representative CNN architectures were evaluated on the binary bird presence-absence detection task. The 50,000-clip dataset was partitioned into training (40,000 clips, 80%), validation (5,000 clips, 10%), and test (5,000 clips, 10%) sets with stratified sampling to ensure balanced class distribution across all splits (50% positive, 50% negative).

All models employ transfer learning from ImageNet-pretrained weights, adapting the final classification layer for binary bird presence detection. Input mel-spectrograms are computed at 224×224 resolution with 224 mel bins, 1024-point FFT, and hop length calculated to achieve exactly 224 time frames over 3-second clips sampled at 16 kHz. While this high-resolution representation exceeds typical embedded deployment constraints, it enables thorough validation using standard computer vision architectures.

Training employs Adam optimizer with cosine learning rate decay (initial rate 1×10^{-4}), batch size 32, and early stopping with patience 15 epochs on validation loss. All models apply ImageNet preprocessing: grayscale mel-spectrograms are replicated to RGB channels, scaled to $[0, 255]$, then processed with architecture-specific normalization. Models were trained on an NVIDIA GTX 1080 Ti GPU using TensorFlow 2.15. All audio was resampled to 16 kHz mono using librosa (v0.10.1) with a Kaiser-windowed sinc resampler. All reported metrics were computed on the held-out test set and averaged across three random seeds (42, 100, 786) to verify stability. Complete experimental code is available at <https://github.com/mun3im/mybad/tree/main/validation>.

4 Results

4.1 Dataset composition

The deduplication stage identified 8 perfect duplicate pairs and 5 near-duplicate pairs, removing 13 files (0.03% of the corpus) and yielding 38,481 unique recordings. Applying the diversity-aware balancing algorithm reduced this to a curated subset of 25,000 positive samples while preserving coverage of all 1,677 species (Figure 3). The mean number of samples per species increased to 14.9 and the Gini coefficient decreased from 0.601 to 0.519, corresponding to a 13.7% reduction in inequality.

The final dataset contains 3,553 unique acoustic clusters. The negative corpus contains exactly 25,000 clips drawn from six source datasets, as summarized in Table 2. Table 3 lists the key properties of the complete SEABAD dataset.

Geographic coverage of positive samples is concentrated in five countries (Figure 4): of 18,999 clips falling within the defined Southeast Asia bounding box (latitudes -9.97° to 19.98° , longitudes 95.37° to 124.88°), Malaysia accounts for 43.1%, Thailand 27.4%, Indonesia 22.0%, Singapore 7.2%, and Brunei 0.2%. Of the remaining 6,001 clips, 602 (2.4%) lacked precise coordinates and 5,399 (21.6%) had coordinates

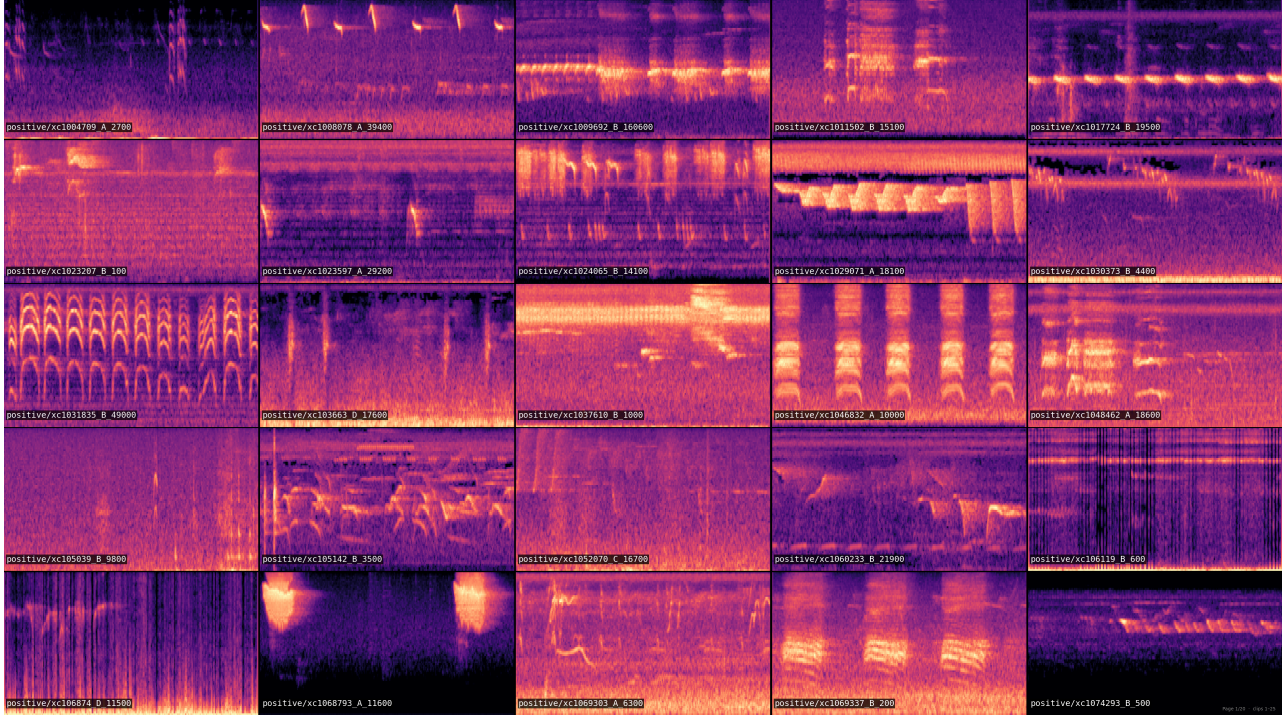


Figure 2: Quality assurance spectrograms showing acoustic diversity across SEABAD positive samples. This 5×5 grid was generated at 4K resolution (3840×2160) for manual auditing on high-resolution displays. Examples show varied bird vocalizations including tonal calls, trills, repeated note sequences, and complex song structures across Southeast Asian species. Spectrograms computed with 80 mel bins, 512 FFT size, 128 hop length, spanning 0–8 kHz over 3-second clips sampled at 16 kHz.

Table 3: SEABAD dataset statistics and characteristics.

Property	Value
Total clips	50,000
Positive (bird presence)	25,000
Negative (bird absence)	25,000
Unique bird species	1,677
Clip duration	3 seconds
Sample rate	16 kHz
Bit depth	16-bit
Format	WAV (16-bit PCM)
Mean samples per species	14.9
Gini coefficient (post-balancing)	0.519
Quality rating A/B	92.1%

Table 4: Geographic distribution of positive samples before and after diversity-aware balancing. Malaysia retains its proportional share (33.1%→33.6%) whereas Thailand decreases (26.4%→24.0%), reflecting prioritization of acoustic diversity and recording quality over uniform geographic sampling.

Country	Pre-balancing		Post-balancing	
	Samples	%	Samples	%
Indonesia	12,996	33.8%	9,155	36.6%
Malaysia	12,743	33.1%	8,400	33.6%
Thailand	10,169	26.4%	5,996	24.0%
Singapore	2,469	6.4%	1,388	5.6%
Brunei	104	0.3%	61	0.2%
Total	38,481	100%	25,000	100%

outside the bounding box. The geographic distribution before and after diversity-aware balancing is shown in Table 4.

4.2 Quality assurance outcomes

The first audit round (500 clips, random seed 42) identified 20 errors; the second round (500 clips, random seed 43) identified 2 additional errors. Of the 1,000 clips inspected, **978 (97.8%)** were judged correct, while **22 (2.2%)** required remediation. The errors were categorized as: wrong onset ($n = 15$), noise dominated ($n = 6$), and no bird ($n = 1$). The most common failure mode was incorrect onset detection, where RMS energy

was dominated by wind or rain bursts preceding the actual vocalization. The combined error rate of 2.2% (22/1,000) yields a final margin of error of $e = 0.90$ percentage points at 95% confidence, confirming dataset labeling accuracy of **97.8% ± 0.9%**. The final dataset has 92.1% of clips rated as quality class A or B with improved mean salience (0.378 vs. 0.367 pre-balancing).

4.3 Baseline validation performance

Table 5 presents performance metrics for four standard architectures and one zero-shot generalist on SEABAD’s test

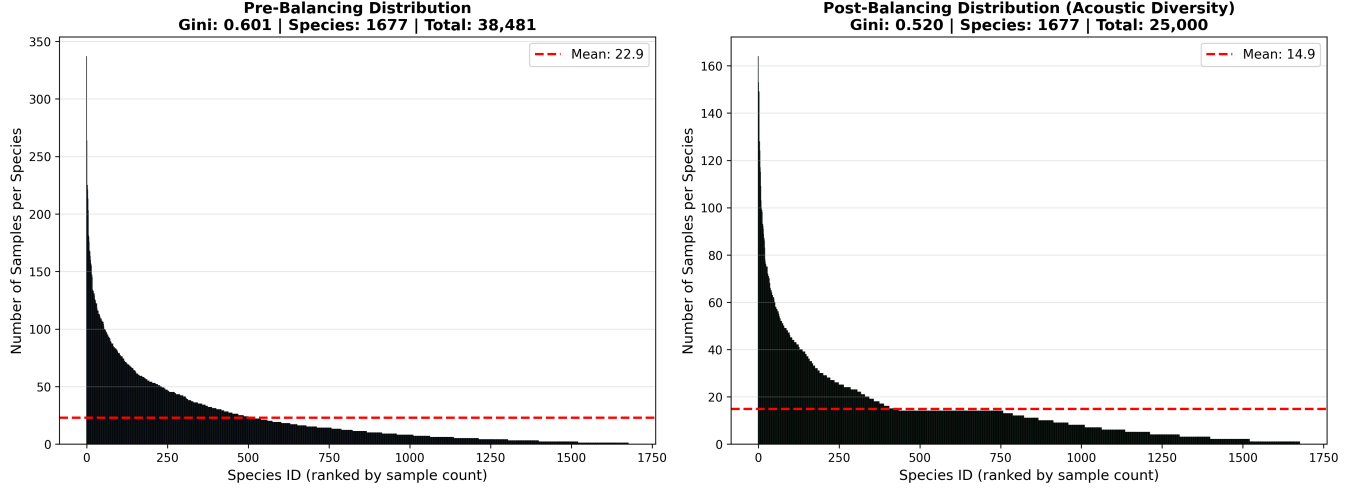


Figure 3: Species distribution before and after diversity-aware balancing. Pre-balancing (left): 38,481 clips across 1,677 species, Gini coefficient 0.601, mean 22.9 clips/species. Post-balancing (right): 25,000 clips preserving all 1,677 species, Gini coefficient 0.519 (13.7% reduction), mean 14.9 clips/species, yielding a more uniform distribution.

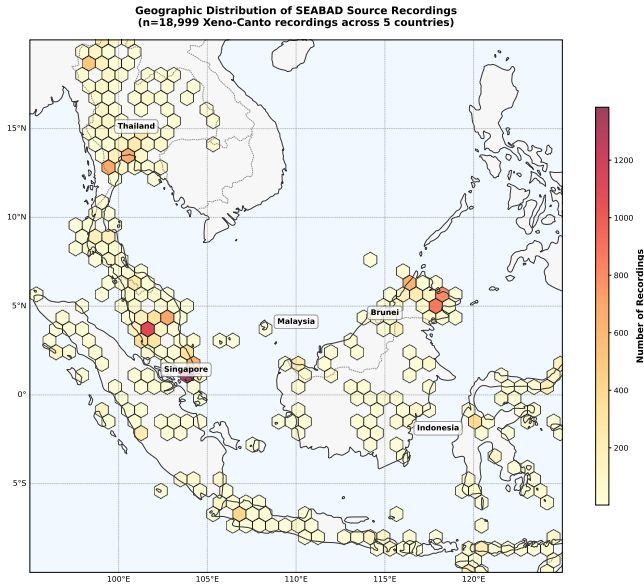


Figure 4: Geographic distribution of SEABAD positive recordings across Southeast Asia. Hexagonal bins show recording density (darker = more recordings). Shown are 18,999 clips within the defined SE Asia bounding box (latitudes -9.97° to 19.98° , longitudes 95.37° to 124.88°), spanning five countries: Malaysia ($n = 8,196$, 43.1%), Thailand ($n = 5,207$, 27.4%), Indonesia ($n = 4,188$, 22.0%), Singapore ($n = 1,376$, 7.2%), and Brunei ($n = 32$, 0.2%). Of the remaining 6,001 clips, 602 (2.4%) lacked precise coordinates and 5,399 (21.6%) had coordinates outside the bounding box. Recordings sourced from the Xeno-Canto community archive.

set. All four trained models achieved exceptional performance across three random seeds: accuracy ranged from 99.49% to 99.73%, AUC from 0.9988 to 0.9996, with seed-to-seed accuracy standard deviation below 0.25% for all architectures.

The mean accuracy across all SEABAD-trained models and seeds was 99.60%.

MobileNetV3-Small, the primary edge-deployment baseline with 1.1M parameters, achieved $99.57\% \pm 0.25\%$ accuracy and AUC 0.9985 ± 0.0002 . Its balanced metrics — precision 0.9956 ± 0.0012 , recall 0.9957 ± 0.0008 — indicate no systematic class bias. This represents a $22\times$ parameter reduction compared to ResNet50 (24.2M) and $13.5\times$ compared to VGG16 (14.9M), with a performance gap of only 0.16% accuracy relative to the best model.

BirdNET v2.4 (Kahl et al. 2021b), a species classification model trained on a global corpus covering roughly 6,000 bird species, was evaluated zero-shot on the SEABAD test set. Predictions were converted to binary bird-activity labels by thresholding the maximum per-clip species confidence at BirdNET’s default threshold $\tau = 0.1$. BirdNET achieved 68.62% accuracy and AUC 0.7819, corresponding to a gap of 30.95 percentage points in accuracy and 0.2166 in AUC relative to MobileNetV3-Small trained on SEABAD.

5 Discussion

5.1 Dataset quality

The consistently strong and stable performance across all four architectures — exceeding 99.4% accuracy and 0.998 AUC across all models and seeds — indicates that SEABAD provides reliable training data with few annotation inconsistencies. The binary detection task exhibits clear acoustic distinction: bird calls produce tonal, harmonic structures with distinct temporal modulation in mel-spectrograms, while the diverse negative class predominantly exhibits broadband, non-tonal characteristics. This acoustic separability, combined with the 97.8% labeling accuracy confirmed by manual audit, confirms that the automated

Table 5: Validation results on SEABAD test set ($n = 5,000$). SEABAD-trained CNNs are averaged across three random seeds (42, 100, 786). BirdNET v2.4 is evaluated zero-shot at threshold $\tau = 0.1$ (BirdNET default). The 30.95 percentage-point accuracy gap between BirdNET and MobileNetV3-Small quantifies the domain-shift cost of applying a temperate-trained generalist to Southeast Asian soundscapes. MobileNetV3-Small serves as the primary edge-deployment baseline.

Model	Params	Accuracy	AUC	Precision	Recall	F1
MobileNetV3-Small [†]	1.1M	99.57 $\pm$ 0.25%	0.9985 $\pm$ 0.0002	0.9956 $\pm$ 0.0012	0.9957 $\pm$ 0.0008	0.9957 $\pm$ 0.0025
EfficientNetB0	4.4M	99.49 $\pm$ 0.23%	0.9991 $\pm$ 0.0004	0.9959 $\pm$ 0.0018	0.9939 $\pm$ 0.0051	0.9949 $\pm$ 0.0023
VGG16	14.9M	99.61 $\pm$ 0.03%	0.9995 $\pm$ 0.0001	0.9960 $\pm$ 0.0014	0.9963 $\pm$ 0.0010	0.9961 $\pm$ 0.0025
ResNet50	24.2M	99.73 $\pm$ 0.02%	0.9992 $\pm$ 0.0003	0.9965 $\pm$ 0.0013	0.9980 $\pm$ 0.0012	0.9973 $\pm$ 0.0019
BirdNET v2.4 [‡]	6.5M	68.62%	0.7819	0.6499	0.8072	0.7201

[†]Primary baseline for edge deployment (intended use case).
[‡]Zero-shot at threshold $\tau = 0.1$ (BirdNET default).

curation pipeline produces high-quality training data without requiring fully manual annotation at scale.

The acoustic salience score can serve as a lightweight proxy to flag likely onset errors before manual review. Clips where RMS energy is dominated by broadband transients (wind, rain bursts) rather than tonal content receive systematically low salience scores. In practice, applying a salience threshold (e.g., retaining clips in the top 80th percentile) substantially reduces the proportion of onset errors in the candidate pool before auditing, lowering manual review burden without sacrificing true-positive recall.

5.2 Domain shift and generalist models

The 30.95 percentage-point accuracy gap between BirdNET and MobileNetV3-Small trained on SEABAD quantifies the domain-shift cost of applying a temperate-trained generalist to Southeast Asian tropical soundscapes. BirdNET’s training corpus contains limited representation of acoustic conditions common in tropical ecosystems, including dense insect biophony, overlapping multi-species choruses, and region-specific vocalization patterns. These results support the central motivation for SEABAD: robust bird activity detection in tropical environments depends heavily on region-specific training data, not only on model architecture or scale.

5.3 Edge deployment context

While MobileNetV3-Small achieves outstanding accuracy, its 1.1M parameters and approximately 4.4MB memory footprint (FP32 weights) exceed AudioMoth’s 256 kB RAM constraint by 17 $\times$. This quantifies the fundamental challenge for microcontroller-based bioacoustic monitoring: achieving high accuracy within extreme constraints (<100K parameters, <256 kB RAM). Initial work addressing this challenge directly on SEABAD (Zabidi 2026c) achieves AUC 0.9840 with only 979 parameters (6.8 kB INT8) and 0.10 ms inference latency on ARM Cortex-M4 hardware, confirming that true microcontroller-class deployment is feasible.

5.4 Methodology transferability

A central goal of this work is transferability. The dual-branch curation pipeline — positive-label stages covering acquisition, deduplication, segment extraction, species balancing, and QA; and negative-label extractions from six heterogeneous sources — has been explicitly designed to generalize beyond Southeast Asian avifauna.

Acoustic similarity-based deduplication. By leveraging FAISS-based nearest-neighbor search over mel spectrogram embeddings, redundant recordings are systematically removed, mitigating train-test leakage.

Diversity-aware species balancing. Stratified sampling combined with acoustic clustering promotes representative coverage across species while limiting over-representation from highly redundant recordings, directly addressing the long-tail distribution characteristic of biodiversity data.

Multi-source negative curation. The integration of environmental sounds, curated datasets, and real-world soundscapes produces negative samples that more closely resemble deployment conditions, enhancing model robustness to both ecological and anthropogenic noise.

These methods are domain-agnostic and can be applied to other tropical regions, temperate ecosystems, or non-avian taxa with minimal modification.

5.5 Implications for tropical PAM

SEABAD has direct implications for PAM in tropical environments. By providing region-specific training data, SEABAD reduces the domain shift caused by differences in species composition, acoustic structure, and background biophony between temperate and tropical ecosystems. More broadly, SEABAD supports the development of cascaded PAM systems in which a lightweight bird activity detector acts as a front-end trigger for more computationally expensive species-level classifiers. Such architectures can substantially reduce energy consumption and processing overhead, extending the operational lifetime of ARUs during long-term deployments.

The strong baseline performance (>99.6% accuracy) should be interpreted in the context of the dataset design. Positive

clips were centered on high-energy vocal segments using RMS-based extraction, while negative samples were drawn from curated non-bird corpora. This reduces ambiguous boundary cases and produces cleaner training examples than those typically encountered in continuous field recordings. In practical deployments, models must still handle more difficult conditions such as overlapping species, distant vocalizations, environmental noise, and degraded recording quality.

5.6 Limitations and future directions

Although SEABAD addresses several gaps in tropical bird activity detection, important limitations remain. The dataset is focused on Southeast Asian ecosystems and may not generalize well to other tropical regions such as the Amazon, Central Africa, or Oceania without additional adaptation. It also relies heavily on community-contributed recordings from Xeno-Canto, which introduce geographic and temporal biases: recordings are often concentrated near accessible locations, biased toward active breeding periods, and more likely to contain distinctive vocalizations.

The composition of the negative corpus is a further limitation. Current negative samples are primarily derived from temperate or globally sourced datasets, leaving some acoustic conditions common in Southeast Asian environments underrepresented, particularly heavy monsoon rainfall, dense cicada choruses, and primate vocalizations. These sounds may contribute to false positives during real-world deployments. SEABAD is designed specifically for binary bird presence-absence detection; species-level classification is outside the scope of this dataset and is not an intended direction.

Several directions for future work remain open: expanding the negative corpus with tropical-specific environmental sounds, conducting long-term field evaluations under varying environmental conditions, systematic error analysis, benchmarking against generalist systems such as Perch, and evaluating transfer across geographic regions. Embedded deployment validation on ARM Cortex-M hardware is addressed in companion work (Zabidi 2026c).

5.7 Conclusions

SEABAD establishes the first large-scale bird activity detection dataset specifically designed for Southeast Asian tropical ecosystems, comprising 50,000 automatically curated 3-second clips spanning over 1,600 species. Beyond releasing this resource, we provide a complete, reproducible dual-branch curation methodology addressing key challenges in biodiversity dataset construction: acoustic deduplication to prevent train-test contamination, diversity-aware species balancing to mitigate long-tail distributions, and multi-source negative curation to ensure robust generalization across deployment conditions.

Baseline validation with MobileNetV3-Small (99.57% $\pm$ 0.25% accuracy, 1.1M parameters) and three larger architectures (>99.4% across all models and seeds) demonstrates SEABAD’s quality with minimal variance,

confirming that the binary detection task is highly learnable. Companion work (Zabidi 2026c) further demonstrates that SEABAD supports training of ultra-lightweight CNNs with under 1,000 parameters (6.8 kB INT8) achieving AUC 0.9840, validating the dataset’s utility for true microcontroller-class deployment.

By releasing SEABAD as an open-access resource alongside detailed curation protocols, we aim to lower barriers for conservation practitioners and accelerate development of autonomous, energy-efficient passive acoustic monitoring systems in biodiversity hotspots worldwide.

Data availability

The SEABAD dataset is publicly available on Zenodo (Zabidi 2026a) with complete documentation and metadata. The curation pipeline and experimental code are available as open-source software (Zabidi 2026b):

- **Dataset:** <https://zenodo.org/records/18290494>
- **Curation scripts:** <https://github.com/mun3im/seabad>

The release includes 50,000 3-second WAV clips (16-bit PCM, 16 kHz mono) with train/validation/test partitions, XC recording ID mappings, preprocessing and deduplication scripts, diversity-aware balancing algorithms, trained baseline model weights, and complete provenance metadata including Xeno-Canto IDs, coordinates, and licenses. All recordings are sourced from Xeno-Canto under Creative Commons licenses (CC BY-SA, CC BY-NC-SA, CC BY-NC-ND, CC0). Users of SEABAD must adhere to the respective Creative Commons licenses when using or redistributing the dataset. The curation pipeline is implemented in Python 3.10+ using librosa, numpy, scikit-learn, and FAISS.

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References

- Colby Banbury, Chuteng Zhou, Igor Fedorov, Ramon Matas, Urmish Thakker, Dibakar Gope, Vijay Janapa Reddi, Matthew Mattina, and Paul Whatmough. Micronets: Neural network architectures for deploying TinyML applications on commodity microcontrollers. *Proceedings of machine learning and systems*, 3:517–532, 2021.
- Meelan Bandara, Roshinie Jayasundara, Isuru Ariyaratne, Dulani Meedeniya, and Charith Perera. Forest sound classification dataset: FSC22. *Sensors*, 23(4):2032, 2023.
- Donald J Benkendorf, Samuel D Schwartz, D Richard Cutler, and Charles P Hawkins. Correcting for the effects of class

- imbalance improves the performance of machine-learning based species distribution models. *Ecological Modelling*, 483:110414, 2023.
- Gerard Bota, Robert Manzano-Rubio, Lidia Catalán, Julia Gómez-Catasús, and Cristian Pérez-Granados. Hearing to the unseen: AudioMoth and BirdNET as a cheap and easy method for monitoring cryptic bird species. *Sensors*, 23(16):7176, 2023.
- Z Burivalova, TM Maeda, Y Rayadin, T Boucher, P Choksi, P Roe, A Truskinger, ET Game, et al. Loss of temporal structure of tropical soundscapes with intensifying land use in borneo. *Science of the Total Environment*, 852:158268, 2022.
- Yohann Chauvier, Niklaus E Zimmermann, Giovanni Poggiato, Daria Bystrova, Philipp Brun, and Wilfried Thuiller. Novel methods to correct for observer and sampling bias in presence-only species distribution models. *Global Ecology and Biogeography*, 30:2312–2325, 2021.
- Nitesh V Chawla, Kevin W Bowyer, Lawrence O Hall, and W Philip Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002.
- Stefano Ciapponi, Leonardo Mannini, Jarek Scanferla, Matteo Anderle, and Elisabetta Farella. Enabling multi-species bird classification on low-power bioacoustic loggers. *arXiv preprint arXiv:2509.20103*, 2025.
- William G. Cochran. *Sampling Techniques*. John Wiley & Sons, 3 edition, 1977.
- Yarden Cohen, David A. Nicholson, Adriana Sanchioni, David J. Perkel, Samuel J. Sober, and Timothy J. Gardner. Tweetynet: A neural network that learns to segment and label birdsong and other temporal patterns. *Methods in Ecology and Evolution*, 13(10):2213–2227, 2022. doi: 10.1111/2041-210X.13928.
- Anne-Sophie Crunchant, Chanakya Dev Nakka, Jason T Isaacs, and Alex K Piel. Acoustic sensors. *Conservation technology*, 53, 2021.
- Bruno da Silva, Axel W. Happi, An Braeken, and Abdellah Touhafi. Evaluation of classical machine learning techniques towards urban sound recognition on embedded systems. *Applied Sciences*, 2019. doi: 10.3390/app9183885.
- Aisling J Daly, Jan M Baetens, and Bernard De Baets. Ecological diversity: measuring the unmeasurable. *Mathematics*, 6(7):119, 2018.
- Robert David, Jared Duke, Advait Jain, Vijay Janapa Reddi, Nat Jeffries, Jian Li, Nick Kreeger, Ian Nappier, Meghna Natraj, Tiezhen Wang, et al. Tensorflow Lite Micro: Embedded machine learning for TinyML systems. *Proceedings of machine learning and systems*, 3:800–811, 2021.
- DCASE Community. DCASE 2018 Task 3: Bird audio detection. <https://dcase.community/challenge2018/task-bird-audio-detection>, 2018. Accessed 2025-12-29.
- Eduardo Fonseca, Jordi Pons Puig, Xavier Favory, Frederic Font Corbera, Dmitry Bogdanov, Andres Ferraro, Sergio Oramas, Alastair Porter, Xavier Serra, Jordi Pons, Xavier Favory, Frederic Font, Dmitry Bogdanov, Andres Ferraro, Sergio Oramas, Alastair Porter, and Xavier Serra. Freesound datasets: a platform for the creation of open audio datasets. *ISMIR, International Society for Music Information Retrieval Conference*, pages 486–493, 2017.
- Sara Fraixedas, Andreas Lindén, Markus Piha, Mar Cabeza, Richard Gregory, and Aleksi Lehtikinen. A state-of-the-art review on birds as indicators of biodiversity: Advances, challenges, and future directions. *Ecological Indicators*, 118:106728, 2020.
- Anthony Gibbons, Ian Donohue, and Andrew Parnell. Monitoring with machines: A review of computational bioacoustics. In Luciano Martins Verdade and Rafael de Andrade Moral, editors, *Wildlife Monitoring: Integrating Conservation and Innovation in Human-Altered Landscapes*, pages 351–368. Springer Nature Switzerland, Cham, 2026. ISBN 978-3-032-05821-8. doi: 10.1007/978-3-032-05821-8_16.
- Corrado Gini. Concentration and dependency ratios. *Rivista di Politica Economica*, 87:769–789, 1909. English translation of Gini’s seminal work on measures of inequality.
- Giovanni Maria Giorgi and Chiara Gigliarano. The Gini concentration index: a review of the inference literature. *Journal of Economic Surveys*, 31(5):1510–1529, 2016.
- Haibo He and Edwardo A Garcia. Learning from imbalanced data. *IEEE Transactions on Knowledge and Data Engineering*, 21(9):1263–1284, 2009.
- Andrew P Hill, Peter Prince, Jake L Snaddon, C Patrick Doncaster, and Alex Rogers. AudioMoth: A low-cost acoustic device for monitoring biodiversity and the environment. *HardwareX*, 6:e00073, 2019. ISSN 24680672. doi: 10.1016/j.ohx.2019.e00073.
- Zhaolan Huang, Adrien Tousnakhoff, Polina Kozyr, Roman Rehausen, Felix Bießmann, Robert Lachlan, Cedric Adjih, and Emmanuel Baccelli. TinyChirp: Bird song recognition using TinyML models on low-power wireless acoustic sensors. In *2024 IEEE 5th International Symposium on the Internet of Sounds (IS2)*, pages 1–10, 2024.
- Alice C. Hughes, Michael C. Orr, Keping Ma, Mark J. Costello, John Waller, Pieter Provoost, Qinmin Yang, Chaodong Zhu, and Huijie Qiao. Sampling biases shape our view of the natural world. *Ecography*, 44(9):1259–1269, 2021. doi: <https://doi.org/10.1111/ecog.05926>.
- Alienor Jeliakov, Yoni Gavish, Charles J Marsh, Jonas Geschke, Neil Brummitt, Duccio Rocchini, Peter Haase, William E Kunin, and Klaus Henle. Sampling and modelling rare species: Conceptual guidelines for the neglected majority. *Global change biology*, 28(12):3754–3777, 2022.
- Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547, 2019.

- Alexis Joly, Hervé Goëau, Christophe Botella, Hervé Glotin, Pierre Bonnet, Willem-Pier Vellinga, Robert Planqué, and Henning Müller. Overview of LifeCLEF 2018: a large-scale evaluation of species identification and recommendation algorithms in the era of AI. In *International Conference of the Cross-Language Evaluation Forum for European Languages*, pages 247–266. Springer, 2018.
- Stefan Kahl, Tom Denton, Holger Klinck, Hervé Glotin, and Hervé Goëau. Overview of BirdCLEF 2021: Bird call identification in soundscape recordings. In *Working Notes of CLEF 2021 – Conference and Labs of the Evaluation Forum*. CEUR-WS, 2021a.
- Stefan Kahl, Connor M Wood, Maximilian Eibl, and Holger Klinck. BirdNET: A deep learning solution for avian diversity monitoring. *Ecological Informatics*, 61:101236, 2021b.
- Chung-Il Kim, Yongjang Cho, Seungwon Jung, Jehyeok Rew, and Eenjun Hwang. Animal sounds classification scheme based on multi-feature network with mixed datasets. *KSII Transactions on Internet and Information Systems*, 14(8): 3384–3398, 2020. doi: 10.3837/tiis.2020.08.013.
- Vincent Lostanlen, Justin Salamon, Mark Cartwright, Brian McFee, Andrew Farnsworth, Steve Kelling, and Juan Pablo Bello. BirdVox-DCASE-20k: Chirp detection in the presence of environmental sounds. In *DCASE Workshop*, April 2018.
- Vincent Lostanlen, Justin Salamon, Andrew Farnsworth, Steve Kelling, and Juan Pablo Bello. Robust sound event detection in bioacoustic sensor networks. *PloS one*, 14(10):e0214168, 2019.
- Vincent Lostanlen, Aurora Cramer, Justin Salamon, Andrew Farnsworth, Benjamin M. Van Doren, Steve Kelling, and Juan Pablo Bello. BirdVoxDetect: Large-scale detection and classification of flight calls for bird migration monitoring. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 32:4134–4145, 2024. doi: 10.1109/TASLP.2024.3444007.
- Isabella Melo, Diego Llusia, Rogério Pereira Bastos, and Luciana Signorelli. Active or passive acoustic monitoring? assessing methods to track anuran communities in tropical savanna wetlands. *Ecological Indicators*, 132:108305, 2021.
- Angela M Mendoza-Henao, Orlando Acevedo-Charry, Daniela Martínez-Medina, Eliana Barona-Cortés, Sergio Córdoba-Córdoba, Paula Caycedo-Rosales, Juan Sebastián Ulloa, Kevin G Borja-Acosta, Alexandra Buitrago-Cardona, and Hoover Pantoja-Sánchez. Past, present, and future of a tropical sounds collection from Colombia. *Bioacoustics*, 32(4):474–490, 2023.
- D. J. Mennill. Field tests of small autonomous recording units: An evaluation of in-person versus automated point counts and a comparison of recording quality. *Bioacoustics*, 33(2): 157–177, 2024. ISSN 21650586. doi: 10.1080/09524622.2024.2315054.
- Carsten Meyer, Patrick Weigelt, and Holger Kreft. Multidimensional biases, gaps and uncertainties in global plant occurrence information. *Ecology Letters*, 19(8):992–1006, 2016. doi: <https://doi.org/10.1111/ele.12624>.
- Veronica Morfi and Dan Stowell. NIPS4Bplus: A richly annotated birdsong audio dataset. In *Proceedings of the Detection and Classification of Acoustic Scenes and Events Workshop (DCASE)*, pages 238–242, 2018.
- Leah Mutanu, Jeet Gohil, Khushi Gupta, Perpetua Wagio, and Gerald Kotonya. A review of automated bioacoustics and general acoustics classification research. *Sensors*, 22(21), 11 2022. ISSN 14248220. doi: 10.3390/s22218361.
- Curtis G. Northcutt, Anish Athalye, and Jonas Mueller. Pervasive label errors in test sets destabilize machine learning benchmarks. In *Proceedings of the 35th Conference on Neural Information Processing Systems, Datasets and Benchmarks Track*, 2021.
- Cristian Pérez-Granados and Juan Traba. Estimating bird density using passive acoustic monitoring: a review of methods and suggestions for further research. *Ibis*, 163(3):765–783, 2021.
- Cristian Pérez-Granados, Gerard Bota, David Giralt, Josep Albarracín, and Juan Traba. Cost-effectiveness assessment of five audio recording systems for wildlife monitoring: differences between recording distances and singing direction. *Ardeola*, 66(2):311–325, 2019.
- Karol J Piczak. ESC: Dataset for environmental sound classification. In *Proceedings of the 23rd ACM international conference on Multimedia*, pages 1015–1018, 2015.
- Ilyas Potamitis, Stavros Ntalampiras, Olaf Jahn, and Klaus Riede. Automatic bird sound detection in long real-field recordings: Applications and tools. *Applied Acoustics*, 80: 1–9, 2014.
- Nirosha Priyadarshani, Stephen Marsland, and Isabel Castro. Automated birdsong recognition in complex acoustic environments: a review. *Journal of Avian Biology*, 49(5): jav-01447, 2018.
- Lukas Rauch, Raphael Schwinger, Moritz Wirth, René Heinrich, Denis Huseljic, Marek Herde, Jonas Lange, Stefan Kahl, Bernhard Sick, Sven Tomforde, et al. BirdSet: A large-scale dataset for audio classification in avian bioacoustics. In *International Conference on Learning Representations*, volume 2025, pages 29482–29520, 2025.
- Samuel RP-J Ross, Darren P O’Connell, Jessica L Deichmann, Camille Desjonquères, Amandine Gasc, Jennifer N Phillips, Sarab S Sethi, Connor M Wood, and Zuzana Burivalova. Passive acoustic monitoring provides a fresh perspective on fundamental ecological questions. *Functional Ecology*, 37(4):959–975, 2023.
- Shravan S. Sethi, Robert M. Ewers, Nicholas S. Jones, Jan Sleutel, and Amira Shabrani. Soundscapes predict species occurrence in tropical forests. *Oikos*, 2022. doi: 10.1111/oik.08525.
- Alexandru-Marius Solomes and Dan Stowell. Efficient bird sound detection on the Bela embedded system. In *ICASSP 2020 – IEEE International Conference on Acoustics, Speech and Signal Processing*, pages 746–750, 2020.

- Dan Stowell. Computational bioacoustics with deep learning: a review and roadmap. *PeerJ*, 10:e13152, 2022.
- Dan Stowell and Mark D. Plumbley. An open dataset for research on audio field recording archives: Freefield1010. *Proceedings of the AES International Conference*, pages 80–86, 2014.
- Dan Stowell, Michael D. Wood, Hanna Pamula, Yannis Stylianou, and Herve Glotin. Automatic acoustic detection of birds through deep learning: the first Bird Audio Detection challenge. *Methods in Ecology and Evolution*, 10(3):368–380, 2018. doi: 10.1111/2041-210X.13103.
- Dan Stowell, Tereza Petruskova, Martin Salek, and Pavel Linhart. Automatic acoustic identification of individuals in multiple species: Improving identification across recording conditions. *Journal of the Royal Society Interface*, 16(153), 4 2019. ISSN 17425662. doi: 10.1098/rsif.2018.0940.
- Dominika Winiarska, Paweł Szymanski, and Tomasz S. Osiejuk. Detection ranges of forest bird vocalisations: Guidelines for passive acoustic monitoring. *Scientific Reports*, 14(1), 12 2024. ISSN 20452322. doi: 10.1038/s41598-024-51297-z.
- Lieven Wittebolle, Massimo Marzorati, Lutz Clement, Alessandra Balloi, Danilo Daffonchio, Kim Heylen, Paul De Vos, Willy Verstraete, and Nico Boon. Initial community evenness favours functionality under selective stress. *Nature*, 458(7238):623–626, 2009.
- Muhammad Mun'im Ahmad Zabidi. SEABAD: Southeast Asia bird activity detection dataset. Zenodo dataset repository, 2026a. URL <https://zenodo.org/records/18290494>.
- Muhammad Mun'im Ahmad Zabidi. SEABAD: Dataset curation and model training scripts. GitHub repository, 2026b. URL <https://github.com/mun3im/seabad>. Open-source implementation of dataset curation pipeline and reproducible training scripts.
- Muhammad Mun'im Ahmad Zabidi. SEABADNet: Lightweight CNNs for tropical bird audio detection on edge devices. GitHub repository, 2026c. URL <https://github.com/mun3im/seabadnet>. Open-source implementation of lightweight CNN architectures for bird audio detection on edge devices.
- Zhao Zhao, Sai-hua Zhang, Zhi-yong Xu, Kristen Bellisario, Nian-hua Dai, Hichem Omrani, and Bryan C Pijanowski. Automated bird acoustic event detection and robust species classification. *Ecological Informatics*, 39:99–108, 2017.